

# National University of Computer and Emerging Sciences, Lahore Campus



**Course:**  
**Program:**  
**Submission**  
**Date:**  
**Instructor:**

**Calculus & Analytical Geometry**  
**CS**

**20-October-22**  
**Dr. Saeeda Zia**

**Exam:**

**Assignment-3**

**Course Code:** MT1003  
**Semester:** Fall 2022

**Total Marks:** 40

**Page(s):** 1

Q1: Find the largest intervals on which  $f$  is (a) Increasing (b) Decreasing (c) Concave up (d) Concave down

$$(a) f(x) = \begin{cases} \frac{1}{2}x^2, & x \leq 0 \\ -x^2, & x > 0 \end{cases}$$

$$(b) f(x) = \frac{x^2}{1-x}$$

Q2: Use the step of the graphing procedure to graph the equation. Include the coordinates of any local absolute extreme points and inflection points

$$(a) Y = \frac{\sqrt{1-x^2}}{2x+1}$$

$$(b) Y = x^2(2-x)$$

Q3(a): Find  $\frac{dy}{dx}$  by implicit differentiation.

$$Y = \sqrt{1 + \sin 3(xy^2)}$$

Q3(b): Find  $\frac{dy}{dx}$

$$Y = \sqrt{\frac{x^2+1}{x^2-5}}$$

Q4(a): An offshore oil well is located in the ocean at a point W, which is 5 mi from the closest shore point A on straight shoreline. The oil is to be piped to a shore point B that is 8 mi from A by piping it on a straight line under water from W to some shore point P between A and B via a pipe along the shoreline. If the cost of laying pipe is \$100,000 per mile under water and \$75,000 per mile over land, where should the point P be located to minimize the cost of laying the pipe?

Q4(b): **Designing a can** what are the dimension of the lightest open-top right circular cylindrical can that will hold a volume of  $1000 \text{ cm}^3$ ?